· Differentiating Through Quaternions

(use q for generalized coordinates, Q for quaternions)

$$\delta Q = \begin{bmatrix} \cos(\frac{\delta\theta}{2}) \\ \hat{\omega}_{/B} \sin(\frac{\delta\theta}{2}) \end{bmatrix} \approx \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{2}\hat{\omega}_{/B}\,\delta\theta \end{bmatrix}$$

$$Q * \delta Q \approx Q + Q * \begin{bmatrix} 0 \\ \frac{1}{2}\hat{\omega}_{/B}\,\delta\theta \end{bmatrix}$$

$$\dot{Q} = Q * \begin{bmatrix} 0 \\ \frac{1}{2}\omega_{/B} \end{bmatrix} = \frac{1}{2} L(Q) H \omega_{/B} \qquad H = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

① $f: Q \to \mathbb{R}^n$ $\quad \delta\phi$ (axis-angle) $\Rightarrow \delta Q \xrightarrow{f} \delta x$

$$f(Q * \delta Q) \approx f\left(Q + Q * \begin{bmatrix} 0 \\ \frac{1}{2}\delta\phi \end{bmatrix}\right)$$
$$= f\left(Q + \tfrac{1}{2} L(Q) H \delta\phi\right)$$
$$\approx f(Q) + \tfrac{1}{2} \frac{\partial f}{\partial x} L(Q) H \delta\phi$$

$G(Q)$: Attitude Jacobian

$$x = f(Q) \qquad \delta x = \tfrac{1}{2} \frac{\partial f}{\partial x} L(Q) H \delta\phi \xrightarrow{\delta t \to 0} \dot{x} = \tfrac{1}{2} \frac{\partial f}{\partial x} L(Q) H \omega_{/B}$$

② $f: \mathbb{R}^n \to Q$

$$\begin{cases} Q * \delta Q \approx Q + Q * \begin{bmatrix} 0 \\ \frac{1}{2}\delta\phi \end{bmatrix} = Q + \frac{1}{2} L(Q) H \delta\phi \\ f(x + \delta x) \approx f(x) + \frac{\partial f}{\partial x} \delta x \end{cases}$$

$$\tfrac{1}{2} L(Q) H \delta\phi = \frac{\partial f}{\partial x} \delta x$$

$$\delta\phi = \underbrace{2 H^T L(Q)^T}_{G(Q)^+} \frac{\partial f}{\partial x} \delta x$$

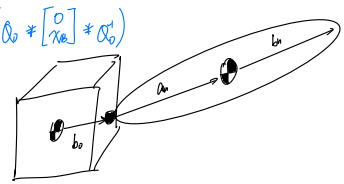
$$Q = f(x) \qquad \delta\phi = 2 H^T L(Q)^T \frac{\partial f}{\partial x} \delta x \xrightarrow{\delta t \to 0} \omega_{/B} = 2 H^T L(Q)^T \frac{\partial f}{\partial x} \dot{x}$$

· Floating - Base Kinematics in 2D

$$R_0 = H^T L(Q_0) R(Q_0)^T H \qquad r_0 \times_{/B} = [0, I]\left(Q_0 * \begin{bmatrix} 0 \\ r_{/B} \end{bmatrix} * Q_0^*\right)$$

$$r_0 = r_0$$

$$\begin{cases} R_n = R_{n-1} \cdot S(\theta_n) \\ r_n = r_{n-1} + R_{n-1} b_{n-1} + R_n a_n \end{cases}$$



$$\text{rotation of each link in quaternion} \leftarrow \begin{bmatrix} r_0 \\ Q_0 \\ r_1 \\ Q_1 \\ \vdots \\ r_n \\ Q_n \end{bmatrix} = K(q) = k\left(\begin{bmatrix} r_0 \\ Q_0 \\ \theta_1 \\ \vdots \\ \theta_n \end{bmatrix}\right) \begin{matrix} \} \text{ base link configuration} \\ \} \text{ joint angle} \end{matrix}$$

$\frac{\partial \delta q}{\partial \delta s}$ (δs is the virtual displacement in V space)

$$\begin{bmatrix} \dot r_0 \\ w_0 \\ \dot r_1 \\ w_1 \\ \vdots \\ \dot r_N \\ w_N \end{bmatrix} = \underbrace{\begin{bmatrix} I & & & \\ & G(Q_0)^T = H^T L(Q_0)^T & & \\ & & I & \\ & & & G(Q_1)^T \\ & & & & \ddots \\ & & & & & I \\ & & & & & & G(Q_N) \end{bmatrix}}_{k(Q)} \underbrace{\frac{\partial k}{\partial q}}_{\dot q} \underbrace{\begin{bmatrix} I & & & \\ & G(Q_0) & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{bmatrix}}_{\frac{\partial k}{\partial q}} \underbrace{\begin{bmatrix} \dot r_0 \\ w_0 \\ \dot\theta_1 \\ \vdots \\ \dot\theta_N \end{bmatrix}}_{V}$$

note that $V \neq \dot q$

M is constant, independent of q

the kinetic energy is

$$T = \sum \frac{1}{2} m_i \|\dot r_i\|^2 + \sum \frac{1}{2} w_i^T J_i w_i = \frac{1}{2} V^T k(q)^T \begin{bmatrix} m_0 I & & & \\ & J_0 & & \\ & & \ddots & \\ & & & m_N I \\ & & & & J_N \end{bmatrix} k(q) V = \frac{1}{2} V^T M(q) V$$

$M(q)$

- Newton-Euler from Lagrangian

assume pure rotation, no translation, no potential

$L = T - U = \frac{1}{2} w^T J w$

$$\min_{w,Q} \int \frac{1}{2} w^T J w \, dt$$
$$\text{s.t. } \dot Q = Q * \begin{bmatrix} 0 \\ \frac{1}{2} w \end{bmatrix} \quad w = 2 H^T L(Q)^T \dot Q$$

$$\partial / \partial \begin{bmatrix} w \\ Q \end{bmatrix} \int \frac{1}{2} w^T J w + l^T (2 H^T L(Q)^T \dot Q - w) \, dt = 0$$

1° $\frac{\partial}{\partial w} \int \frac{1}{2} w^T J w - l^T w \, dt = 0$ $\quad l = Jw$ dual variable is the angular momentum

2° $\frac{\partial}{\partial Q} \int l^T 2 H^T L(Q)^T \dot Q \, dt = 0$

$$\begin{aligned} 0 &= \frac{d}{d\epsilon} \int l^T H^T \left(L(Q) + \epsilon L(\delta Q) \right)^T (\dot Q + \epsilon \dot{\delta Q}) \, dt \Big|_{\epsilon=0} \\ &= \frac{d}{d\epsilon} \int l^T H^T \left(L(Q)^T \dot Q + \epsilon L(\delta Q)^T \dot Q + \epsilon L(Q)^T \dot{\delta Q} + \epsilon^2 L(\delta Q)^T \dot{\delta Q} \right) dt \Big|_{\epsilon=0} \\ &= \int l^T H^T \left(L(\delta Q)^T \dot Q + L(Q)^T \dot{\delta Q} \right) dt \\ &= \int l^T H^T \left(\delta Q^{-1} * \dot Q + Q^{-1} * \dot{\delta Q} \right) dt \quad T = \begin{bmatrix} I & \\ & -I \end{bmatrix} \quad TQ = Q^{-1} \\ &= \int l^T H^T R(Q) T \delta Q + \int l^T H^T (Q^{-1} * \dot{\delta Q}) \, dt \end{aligned}$$

let $\delta\phi$ be the infinitesimal change in axis-angle that generates δQ

$\delta Q = \sum L(Q) H \delta\phi$

$$\begin{aligned} \frac{d}{dt} \delta Q &= \sum L(Q) H \dot{\delta\phi} + \sum \frac{\partial}{\partial Q} (L(Q) H \delta\phi) \dot Q \\ &= \sum L(Q) H \dot{\delta\phi} + \sum \frac{\partial}{\partial Q} (R(H\delta\phi) Q) \dot Q \\ &= \sum L(Q) H \dot{\delta\phi} + \sum \dot Q * (H\delta\phi) \end{aligned}$$

$$= \int l^T H^T R(Q) T \delta Q \, dt + \frac{1}{2} \int l^T H^T (Q^{-1} * L(Q) H \dot{\delta\phi}) \, dt + \frac{1}{2} \int l^T H^T (Q^{-1} * \dot Q * H \delta\phi) \, dt$$

$$= \frac{1}{2}\int l^T H^T R(Q) T L(Q) H \dot{\phi}\, dt + \frac{1}{2}\int l^T H^T L(Q)^T L(Q) H \dot{\phi}\, dt + \frac{1}{2}\int l^T H^T L(Q)^T L(Q) H \dot{\phi}\, dt$$

$$= \frac{1}{2}\int l^T H^T \left(R(Q) T L(Q) H + L(Q)^T L(Q) H \right) \dot{\phi}\, dt - \frac{1}{2}\int \dot{l}^T \delta\phi\, dt$$

$$H^T \left[R(Q) T L(Q) H + L(Q)^T L(Q) H \right] \times$$

$$= H^T \left[(Q * [\delta r]) * \dot{Q} + Q * * \dot{Q} * [\delta r] \right]$$

$$= H^T \left[[\delta r] * Q^{-1} * \dot{Q} + Q^{-1} * \dot{Q} * [\delta r] \right]$$

$$= H^T \left([-\omega] * [\delta r] + [\delta r] * [\omega] \right)$$

$$= \omega \times r = [\omega] r$$

$$= \frac{1}{2}\int l^T [\omega] \dot{\phi}\, dt - \frac{1}{2}\int \dot{l}^T \delta\phi\, dt$$

$$[\omega]^T l - \dot{l} = 0$$

$$[\omega] l + \dot{l} = 0$$

$\omega \times J\omega + J\dot{\omega} = 0$ Newton-Euler Equation

generalize to any Lie Group

· Floating-Base Dynamics

let $\theta = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_N \end{bmatrix}$ $q = \begin{bmatrix} r_0 \\ Q_0 \\ \theta \end{bmatrix}$ $v = \begin{bmatrix} \dot{r}_0 \\ \omega_0 \\ \dot{\theta} \end{bmatrix}$

$$\min_{r_0, Q_0, \omega_0, \theta} \int L(q,v)\, dt = \int \frac{1}{2} v^T M(q) v - U(q)\, dt$$

s.t. $\dot{Q}_0 = Q_0 * \begin{bmatrix} 0 \\ \frac{1}{2}\omega_0 \end{bmatrix}$ $\omega_0 = 2 H^T L(Q_0)^T \dot{Q}_0$

$$\frac{\partial}{\partial \begin{bmatrix} r_0 \\ Q_0 \\ \omega_0 \\ \theta \end{bmatrix}} \int \frac{1}{2} v^T M(q) v - U(q) + l^T \left(2 H^T L(Q_0)^T \dot{Q}_0 - \omega_0 \right) dt$$

1° $\frac{\partial}{\partial \begin{bmatrix} r_0 \\ \theta \end{bmatrix}} \int L(q,v)\, dt = 0$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{r}_0} - \frac{\partial L}{\partial r_0} = 0$ Euler-Lagrange Equation

$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$

2° $\frac{\partial}{\partial \omega_0} \int L(q,v) - l^T \omega_0\, dt$

$l = \frac{\partial L}{\partial \omega_0}$

3° $\frac{\partial}{\partial \delta Q_0} \int L(q,v) + l^T \left(2 H^T L(Q_0)^T \dot{Q}_0 \right) dt$

$$0 = \frac{\partial}{\partial \epsilon} \int L\left(\begin{bmatrix} r_0 \\ Q_0 * \epsilon \delta Q_0 \\ \theta \end{bmatrix}, v \right) + l^T 2 H^T \left((Q_0 + \epsilon \delta Q_0)^{-1} * (\dot{Q}_0 + \epsilon \dot{\delta Q}_0) \right) dt$$

$$= \int \frac{\partial L}{\partial \delta Q_0}^T \delta Q_0\, dt + \int l^T [\omega] \delta\phi\, dt - \int \dot{l}^T \delta\phi\, dt$$

$$= \int \frac{\partial L}{\partial \delta Q_0}^T \frac{1}{2} L(Q_0) H \dot{\delta\phi}\, dt + \int l^T [\omega] \delta\phi - \int \dot{l}^T \delta\phi\, dt$$

$$\frac{1}{2} H^T L(Q_0)^T \frac{\partial L}{\partial \delta Q_0} - [\omega] l - \dot{l} = 0$$

$$\frac{1}{2} H^T L(Q_0)^T \frac{\partial L}{\partial \delta Q_0} - [\omega] \frac{\partial L}{\partial \omega_0} - \frac{d}{dt}\frac{\partial L}{\partial \dot{\delta Q_0}} = 0$$

$$\frac{d}{dt}\frac{\partial L}{\partial \omega_0} + [\omega] \times \frac{\partial L}{\partial \omega_0} - \frac{1}{2} H^T L(Q_0)^T \frac{\partial L}{\partial \delta Q_0} = 0$$

4° Combine

$$\begin{cases} \frac{d}{dt}\left[\frac{\partial L}{\partial \dot r_0}\right] - \frac{\partial L}{\partial r_0} = 0 \\ \frac{d}{dt}\left[\frac{\partial L}{\partial \dot \theta}\right] - \frac{\partial L}{\partial \theta} = 0 \\ \frac{d}{dt}\left[\frac{\partial L}{\partial w_0}\right] + w_0 \times \frac{\partial L}{\partial w_0} - \frac{1}{2} H^T L(Q_0)^T \frac{\partial L}{\partial Q_0} = 0 \end{cases}$$

$$M(q)\begin{bmatrix} \dot r_0 \\ w_0 \\ \ddot\theta \end{bmatrix} = M(q)V$$

$$\frac{d}{dt}\left(M(q)v\right) = M(q)\dot v + \frac{d}{dq}\left(M(q)v\right)\dot q$$
$$= M(q)\dot v + \frac{d}{dq}\left(M(q)v\right)\frac{dq}{dv}v$$

work done $= F_{ext}^T Dr_i(q)\,\delta q = F_{ext}^T Dr_i(q)\frac{dq}{dv}\,\underline{\delta s}$ virtual displacement in v space

Eg $\dot Q_0 = \frac{1}{2} L(Q_0) H w_0 \qquad \delta Q_0 = \frac{1}{2} L(Q_0) H \delta\phi$

$dq/ds = dq/dv$

$$M(q)\begin{bmatrix} \ddot r_0 \\ \dot w_0 \\ \ddot\theta \end{bmatrix} + \frac{\partial}{\partial q}\left(M(q)v\right)\frac{dq}{dv}v + \begin{bmatrix} -\partial L/\partial r_0 \\ w_0 \times \frac{\partial L}{\partial w_0} - \frac{1}{2} H^T L(Q_0)^T \frac{\partial L}{\partial Q_0} \\ -\partial L/\partial \theta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \tau \end{bmatrix} + \sum_i \frac{dq}{dv}^T Dr_i(q)^T F_{i,ext}$$

τ: joint torque

not safe to compute, especially when $r_i(q)$ include quaternion computation

work done by $F_{i,ext}$ is $F_{i,ext}^T \delta r_i = \left[\delta r_{0/E} + (\delta\phi \times (r_i - r_0)_{/E} + \frac{\partial r_i}{\partial\theta}\delta\theta\right]^T F_{i,ext}$

1° translation work $\frac{\partial}{\partial \delta r_0}\left[\delta r_0^T F_{i,ext}\right] = F_{i,ext}$

2° rotational work $\frac{\partial}{\partial \delta\phi}\left[(\delta\phi \times (r_i - r_0)_{/B})^T F_{i,ext}\right] = \frac{\partial}{\partial \delta\phi}\left[(-[(r_i - r_0)_{/B}]\delta\phi)^T F_{i,ext}\right]$
$= -[(r_i - r_0)_{/B}]^T F_{i,ext}$

$$\frac{\partial}{\partial \delta\phi}\left[\left(\delta\phi \times (r_i - r_0)_{/B}\right)_{/E}^T F_{i,ext}\right] = \frac{\partial}{\partial \delta\phi}\left[\left(\delta\phi \times (r_i - r_0)_{/B}\right)^T F_{i,ext/B}\right]$$

$\begin{bmatrix} D \\ F_{/E} \end{bmatrix} = Q_0 * \begin{bmatrix} D \\ F_{/B} \end{bmatrix} * Q_0^{-1}$

$F_{/B} = H^T\left(Q_0^{-1} * H F_{/E} * Q_0\right)$
$= H^T L(Q_0)^T R(Q_0) H F_{/E}$

$$= \frac{\partial}{\partial \delta\phi}\left[\left(-[(r_i - r_0)_{/B}]\delta\phi\right)^T F_{i,ext/B}\right]$$
$$= -[(r_i - r_0)_{/B}]^T F_{i,ext/B}$$
$$= [(r_i - r_0)_{/B}] F_{i,ext/B}$$
$$= [(r_i - r_0)_{/B}] H^T L(Q_0)^T R(Q_0) H \; F_{i,ext}$$

3° joint work $\frac{\partial}{\partial\delta\theta}\left(\left(\frac{\partial r_i}{\partial\theta}\delta\theta\right)^T F_{i,ext}\right) = \frac{\partial r_i}{\partial\theta}^T F_{i,ext}$

$$\begin{bmatrix} 0 \\ 0 \\ \tau \end{bmatrix} + \sum_i \begin{bmatrix} I \\ [(r_i - r_0)_{/B}] H^T L(Q_0)^T R(Q_0) H \\ \frac{\partial r_i}{\partial\theta}^T \end{bmatrix} F_{i,ext}$$

- Differentiate through quaternions

  (use q for generalized coordinates, Q for quaternions)

$$\delta Q = \begin{bmatrix} \cos(\frac{\|\delta\theta\|}{2}) \\ \hat{w}_{/B} \sin(\frac{\|\delta\theta\|}{2}) \end{bmatrix} \approx \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{2} \hat{w}_{/B} \delta\theta \end{bmatrix}$$

$$Q * \delta Q \approx Q + Q * \begin{bmatrix} 0 \\ \frac{1}{2} \hat{w}_{/B} \delta\theta \end{bmatrix}$$

$$\dot{Q} = Q * \begin{bmatrix} 0 \\ \frac{1}{2} w_{/B} \end{bmatrix} = \frac{1}{2} L(Q) H w_{/B} \quad H = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

① $f: Q \mapsto \mathbb{R}^n$ $\quad \delta\phi$ (axis-angle) $\Rightarrow \delta Q \overset{?}{\Rightarrow} \delta x$

$$f(Q * \delta Q) = f(Q + Q * \begin{bmatrix} 0 \\ \frac{1}{2}\delta\phi \end{bmatrix})$$
$$= f(Q + \frac{1}{2} L(Q) H \delta\phi)$$
$$\approx f(Q) + \frac{1}{2} \frac{\partial f}{\partial Q} L(Q) H \delta\phi$$

$G(Q)$: derivate jacobian

$x = f(Q) \quad \delta x \approx \frac{1}{2} \frac{\partial f}{\partial Q} L(Q) H \delta\phi \xrightarrow{\delta t \to 0} \dot{x} = \frac{1}{2} \frac{\partial f}{\partial Q} L(Q) H w_{/B}$

② $f: \mathbb{R}^n \mapsto Q$

$$\begin{cases} Q * \delta Q \approx Q + Q * \begin{bmatrix} 0 \\ \frac{1}{2}\delta\phi \end{bmatrix} = Q + \frac{1}{2} L(Q) H \delta\phi \\ f(x + \delta x) \approx f(x) + \frac{\partial f}{\partial x} \delta x \end{cases}$$
$$\frac{1}{2} L(Q) H \delta\phi \approx \frac{\partial f}{\partial x} \delta x$$
$$\delta\phi \approx 2 \underbrace{H^T L(Q)^T}_{G(Q)^+} \frac{\partial f}{\partial x} \delta x$$

$Q = f(x) \quad \delta\phi \approx 2 H^T L(Q)^T \frac{\partial f}{\partial x} \delta x \xrightarrow{\delta t \to 0} w_{/B} \approx 2 H^T L(Q)^T \frac{\partial f}{\partial x} \dot{x}$

- Floating-Base kinematics in 3D

$$R_0 = H^T L(Q_0) R(Q_0)^T H \qquad R_0 x_{/B} = [0\ I] \left(Q_0 * \begin{bmatrix} 0 \\ x_{/B} \end{bmatrix} * Q_0^{-1} \right)$$

$r_0 = r_0$

$$\begin{cases} R_n = R_{n-1} \cdot S_n(\theta_n) \\ r_n = r_{n-1} + R_{n-1} \cdot b_{n-1} + R_n \cdot Q_n \end{cases}$$

rotation of each link in quaternion

$$\begin{bmatrix} r_0 \\ Q_0 \\ r_1 \\ Q_1 \\ \vdots \\ r_N \\ Q_N \end{bmatrix} = k(q) = k\left(\begin{bmatrix} r_0 \\ Q_0 \\ \theta_1 \\ \vdots \\ \theta_N \end{bmatrix} \right) \begin{matrix} \} \text{ base link configuration} \\ \} \text{ joint angle} \end{matrix}$$

$\frac{\partial q}{\partial s}$ $\delta q = \frac{\partial q}{\partial s}\delta s$ where δs is the infinitesimal displacement in V-space

$$\begin{bmatrix} \dot r_0 \\ w_0 \\ \dot r_1 \\ w_1 \\ \vdots \end{bmatrix} = \begin{bmatrix} I & & & \\ & 2H^T L(q_0)^T & & \\ & & I & \\ & & & 2H^T L(q_1)^T \\ & & & & \ddots \end{bmatrix} \frac{\partial \dot q}{\partial \dot q} \begin{bmatrix} I & & & \\ & \frac{1}{2}L(q_0)H & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{bmatrix} \begin{bmatrix} \dot r_0 \\ w_0 \\ \dot\theta_1 \\ \vdots \\ \dot\theta_N \end{bmatrix}$$

$S(q)$ $\qquad$ V more than $V \neq \dot q$

$\dot q$

M is constant, independent of q

The kinetic Energy is

$$T = \sum_i \frac{1}{2} m_i \|\dot r_i\|^2 + \sum_i \frac{1}{2} w_i^T J_i w_i = \frac{1}{2} V^T S(q)^T \begin{bmatrix} m_1 I & & & \\ & J_1 & & \\ & & \ddots & \\ & & & m_N I \\ & & & & J_N \end{bmatrix} S(q) V = \frac{1}{2} V^T M(q) V$$

$M(q)$

- Newton - Euler from Lagrangian

  assume pure rotation, no translation, no potential

  $L = T - U = \frac{1}{2} w^T J w$

  $\Downarrow$

  $\min_{w,q} \int \frac{1}{2} w^T J w \, dt$

  s.t. $\dot q = q * \begin{bmatrix} 0 \\ \frac{1}{2} w \end{bmatrix}$ $\qquad w = 2H^T L(q)^T \dot q$

  $\frac{d}{d\begin{bmatrix} w \\ q \end{bmatrix}} \int \frac{1}{2} w^T J w + l^T \left(2H^T L(q)^T \dot q - w \right) dt = 0$

  1° $\frac{d}{dw} \int \frac{1}{2} w^T J w - l^T w \, dt = 0 \Rightarrow$ $l = Jw$ dual variable is the angular momentum

  2° $\frac{d}{dq} \int l^T 2H^T L(q)^T \dot q \, dt = 0$

$$\begin{aligned} 0 &= \frac{d}{d\varepsilon} \int l^T H^T \left(L(q) + \varepsilon L(\delta q) \right)^T \left(\dot q + \varepsilon \dot{\delta q} \right) dt \Big|_{\varepsilon=0} \\ &= \frac{d}{d\varepsilon} \int l^T H^T \left(L(q)^T \dot q + \varepsilon L(q)^T \dot{\delta q} + \varepsilon L(\delta q)^T \dot q + \varepsilon^2 L(\delta q)^T \dot{\delta q} \right) dt \Big|_{\varepsilon=0} \\ &= \int l^T H^T \left(L(q)^T \dot{\delta q} + L(\delta q)^T \dot q \right) dt \\ &= \int l^T H^T \left(q^{-1} * \dot{\delta q} + \delta q^{-1} * \dot q \right) dt \qquad T = \begin{bmatrix} 1 & \\ & -I \end{bmatrix} \quad Tq = q^{-1} \\ &= \int l^T H^T (q^{-1} * \dot{\delta q}) \, dt + \int l^T H^T R(q) T \delta q \, dt \end{aligned}$$

  let $\delta\phi$ be the infinitesimal change in axis-angle that generates δq

  $\delta q = \frac{1}{2} L(q) H \delta\phi$

  $\frac{d}{dt}(\delta q) = \frac{1}{2} L(q) H \dot{\delta\phi} + \frac{1}{2} \frac{\partial}{\partial q} \left(L(q) H \delta\phi \right) \dot q$

  $= \frac{1}{2} L(q) H \dot{\delta\phi} + \frac{1}{2} \frac{\partial}{\partial q} \left(R(H\delta\phi) q \right) \dot q$

  $= \frac{1}{2} L(q) H \dot{\delta\phi} + \frac{1}{2} \dot q * (H\delta\phi)$

$$= \int l^T H^T R(q) T \delta q \, dt + \frac{1}{2} \int l^T H^T (q^{-1} * L(q) H \dot{\delta\phi}) \, dt + \frac{1}{2} \int l^T H^T (q^{-1} * \dot q * H\delta\phi) \, dt$$

$$= \frac{1}{2}\int [\dot{l}^T H^T R(\alpha) T L(\alpha) H S\dot{\phi}\, dt + \frac{1}{2}\int [l^T H^T L(\alpha)^T L(\dot{\alpha}) H S\dot{\phi}\, dt + \frac{1}{2}\int [l^T H^T L(\alpha)^T L(\alpha) H S\dot{\phi}\, dt$$

$$= \frac{1}{2}\int [\dot{l}^T H^T \left(R(\alpha) T L(\alpha) H + L(\alpha)^T L(\dot{\alpha}) H\right) S\dot{\phi}\, dt - \frac{1}{2}\int i^T S\dot{\phi}\, dt$$

$$H^T \left(R(\alpha) T L(\alpha) H + L(\alpha)^T L(\dot{\alpha}) H\right) x$$

$$= H^T \left\{ \left(\alpha * \begin{bmatrix} 0 \\ x \end{bmatrix}\right)^{\cdot} * \dot{\alpha} + \alpha^{-1} * \dot{\alpha} * \begin{bmatrix} 0 \\ x \end{bmatrix} \right\}$$

$$= H^T \left\{ \begin{bmatrix} 0 \\ \dot{x} \end{bmatrix} * \alpha^{-1} * \dot{\alpha} + \alpha^{-1} * \dot{\alpha} * \begin{bmatrix} 0 \\ x \end{bmatrix} \right\}$$

$$= H^T \left\{ \begin{bmatrix} 0 \\ \dot{x} \end{bmatrix} * \begin{bmatrix} 0 \\ \frac{1}{2} w_0 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{2} w_0 \end{bmatrix} * \begin{bmatrix} 0 \\ x \end{bmatrix} \right\}$$

$$= w \times x = [w] x$$

$$= \frac{1}{2}\int [l^T [w] S\dot{\phi}\, dt - \frac{1}{2}\int i^T S\dot{\phi}\, dt$$

$$[w]^T l - \dot{l} = 0$$

$$[w] l + \dot{l} = 0$$

$w \times Jw + J\dot{w} = 0$ Newton-Euler Equation

generalize to any lie group

- Floating-Base Dynamics

let $\theta = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_N \end{bmatrix}$ $q = \begin{bmatrix} r_0 \\ \alpha_0 \\ \theta \end{bmatrix}$ $v = \begin{bmatrix} \dot{r}_0 \\ w_0 \\ \dot{\theta} \end{bmatrix}$

$$\min_{r_0\, \alpha_0\, w_0\, \theta} \int L(q,v)\, dt = \frac{1}{2}\int v^T M(q) v - U(q)\, dt$$

$$\text{s.t.} \quad \dot{\alpha}_0 = \alpha_0 * \begin{bmatrix} 0 \\ \frac{1}{2} w_0 \end{bmatrix} \qquad w_0 = 2 H^T L(\alpha_0)^T \dot{\alpha}_0$$

$$\frac{\partial}{\partial \begin{bmatrix} r_0 \\ \alpha_0 \\ \theta \end{bmatrix}} \int \frac{1}{2} v^T M(q) v - U(q) + l^T \left(2 H^T L(\alpha_0)^T \dot{\alpha}_0 - w_0\right) dt = 0$$

1° $\frac{\partial}{\partial \begin{bmatrix} r_0 \\ \theta \end{bmatrix}} \int L(q,v)\, dt = 0$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{r}_0} - \frac{\partial L}{\partial r_0} = 0 \qquad \text{Euler-Lagrange Equation}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$$

2° $\frac{\partial}{\partial w_0} \int L(q,v) - l^T w_0\, dt = 0$

$$l = \frac{\partial L}{\partial w_0}$$

3° $\frac{\partial}{\partial \dot{\alpha}_0} \int L(q,v) + l^T 2 H^T L(\alpha_0)^T \dot{\alpha}_0\, dt$

$$0 = \frac{d}{dt}\int L\left(\begin{bmatrix} r_0 \\ \alpha_0 + \epsilon \delta\alpha_0 \\ \theta \end{bmatrix}, v\right) + 2 l^T H^T \left((\alpha_0 + \epsilon \delta\alpha_0)^{\cdot} * (\alpha_0 + \epsilon \delta\alpha_0)\right) dt$$

$$= \int \frac{\partial L}{\partial \dot{\alpha}_0}^T \delta\dot{\alpha}_0\, dt + \int l^T [w] S\dot{\phi}\, dt - \int \dot{l}^T S\dot{\phi}\, dt$$

$$= \int \frac{\partial L}{\partial \dot{\alpha}_0}^T \frac{1}{2} L(\alpha_0) H S\dot{\phi}\, dt + \int l^T [w] S\dot{\phi}\, dt - \int \dot{l}^T S\dot{\phi}\, dt$$

$$\frac{1}{2} H^T L(\alpha_0)^T \frac{\partial L}{\partial \dot{\alpha}_0} - [w] l - \dot{l} = 0$$

$$\frac{1}{2} H^T L(\alpha_0)^T \frac{\partial L}{\partial \dot{\alpha}_0} - [w] \frac{\partial L}{\partial w_0} - \frac{d}{dt}\frac{\partial L}{\partial w_0} = 0$$

$$\frac{d}{dt}\frac{\partial L}{\partial w_0} + w_0 \times \frac{\partial L}{\partial w_0} - \frac{1}{2} H^T L(\alpha_0)^T \frac{\partial L}{\partial \dot{\alpha}_0} = 0$$

4° Combine

$$\begin{cases} \frac{d}{dt}\boxed{\frac{\partial L}{\partial \dot r_0}} - \frac{\partial L}{\partial r_0} = 0 \\ \frac{d}{dt}\boxed{\frac{\partial L}{\partial \omega_0}} + v_0 \times \frac{\partial L}{\partial \omega_0} - \frac{1}{2} H^T L(\alpha)^T \frac{\partial L}{\partial \alpha_0} = 0 \\ \frac{d}{dt}\boxed{\frac{\partial L}{\partial \dot\theta}} - \frac{\partial L}{\partial \theta} = 0 \end{cases}$$

$$\Downarrow$$

$$M(q)\begin{bmatrix} \dot v_0 \\ \dot\omega_0 \\ \ddot\theta \end{bmatrix} = M(q)V$$

$$\frac{d}{dt}\left(M(q)V\right) = M(q)\dot V + \frac{\partial}{\partial q}\left(M(q)V\right)\dot q$$

$$= M(q)\dot V + \frac{\partial}{\partial q}\left(M(q)V\right)\frac{\partial q}{\partial s}V$$

work done $= F_{i,ext}^T \, Dr_i(q)\,\delta q = F_{i,ext}^T \, Dr_i(q)\frac{\partial q}{\partial s}\delta s$

$$M(q)\begin{bmatrix} \dot v_0 \\ \dot\omega_0 \\ \ddot\theta \end{bmatrix} + \frac{d}{dq}\left(M(q)V\right)\frac{\partial q}{\partial s}V + \begin{bmatrix} -\partial L/\partial r_0 \\ v_0 \times \partial L/\partial \omega_0 - \frac{1}{2}H^T L(\alpha)^T \frac{\partial L}{\partial \alpha_0} \\ -\partial L/\partial \theta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \tau \end{bmatrix} \} \text{ joint pose} + \underbrace{\sum_i \frac{\partial q}{\partial s}^T Dr_i(q)^T F_{i,ext}}$$

not safe to compute, especially when $r_i(q)$ include quaternion computation

work done by $F_{i,ext}$ is $F_{i,ext}^T \delta r_i = \left[\delta r_{0/I} + (\delta\phi \times (r_i - r_0)_{/B})_{/I} + \frac{\partial r_i}{\partial \theta}\delta\theta\right]^T F_{i,ext}$

1° translation work: $\frac{\partial}{\partial \delta r_0}\left[\delta r_0^T F_{i,ext}\right] = F_{i,ext}$

2° rotational work: $\frac{\partial}{\partial \delta\phi}\left[(\delta\phi \times (r_i - r_0)_{/B})_{/I}^T F_{i,ext}\right] = \frac{\partial}{\partial \delta\phi}\left[-\left[(r_i - r_0)_{/B}\right]\delta\phi\right)^T F_{i,ext/B}\Big]$

$$= -\left[(r_i - r_0)_{/B}\right]^T F_{i,ext/B}$$

$$= \left[(r_i - r_0)_{/B}\right] F_{i,ext/B}$$

$$= \left[(r_i - r_0)_{/B}\right] H^T L(\alpha)^T R(\alpha)^T H \, F_{i,ext}$$

3° joint work $\frac{\partial}{\partial \delta\theta}\left(\left(\frac{\partial r_i}{\partial \theta}\delta\theta\right)^T F_{i,ext}\right) = \frac{\partial r_i}{\partial \theta}^T F_{i,ext}$

$$\begin{bmatrix} 0 \\ 0 \\ \tau \end{bmatrix} + \sum_i \underbrace{\begin{bmatrix} I \\ \left[(r_i - r_0)_{/B}\right] H^T L(\alpha)^T R(\alpha) H \\ \frac{\partial r_i}{\partial \theta}^T \end{bmatrix} F_{i,ext}}_{\frac{\partial q}{\partial s}^T Dr_i(q)^T}$$